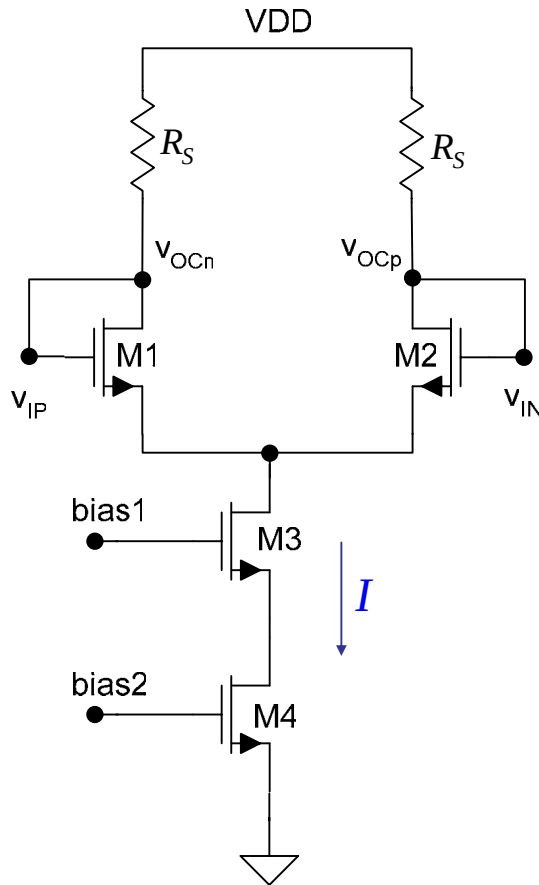


CMFB

Common Mode Feedback

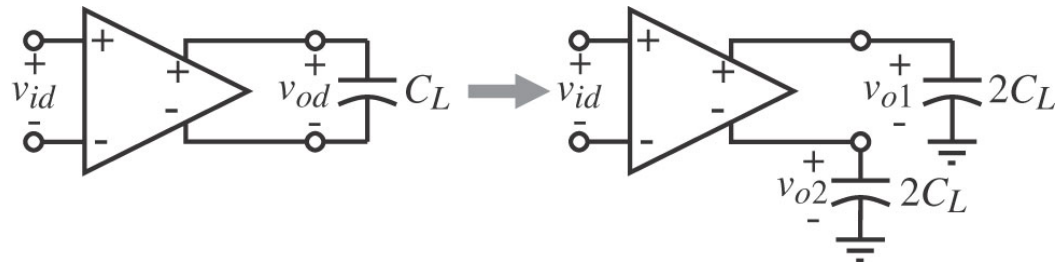
Diffamp with Resistive Pull up



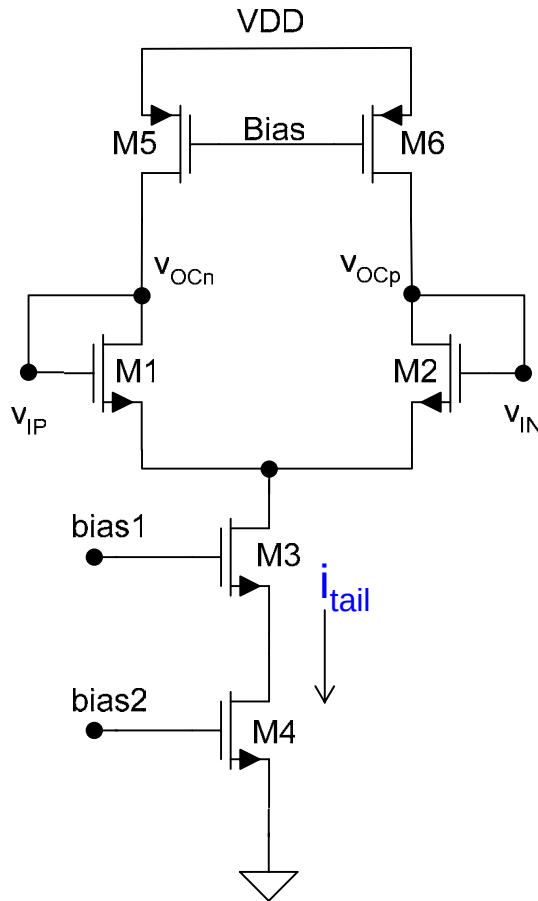
- When inputs and outputs are shorted the DC common-mode is well defined

$$V_{CM} = V_{DD} - \frac{1}{2} I R_S$$

- Effective diagrams when driving capacitive loads



Active Load Issues



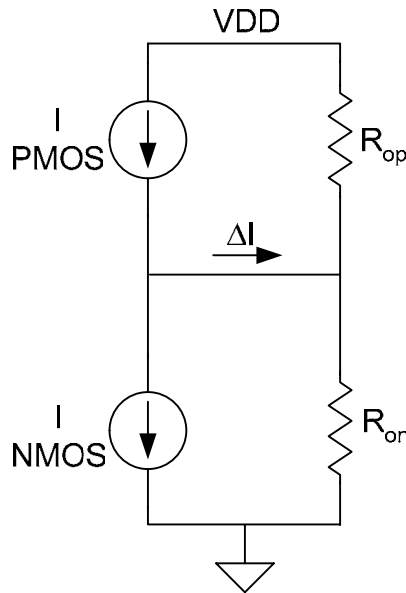
- The differential gain increases with active loads

- ↳ However the common-mode bias point suffers
- ↳ It is difficult to match a P source to an N sink
- ↳ Cannot set DC common mode with differential feedback

1. Assume M5 and M6 each carry a current greater than i_{tail}
2. The current at the drain of M3 must add up to i_{tail}
3. The only way to accommodate the decrease in current is by pushing M5,6 to triode
4. For the reverse case M3,4 will enter triode to accommodate the lower current being sourced by M5,6

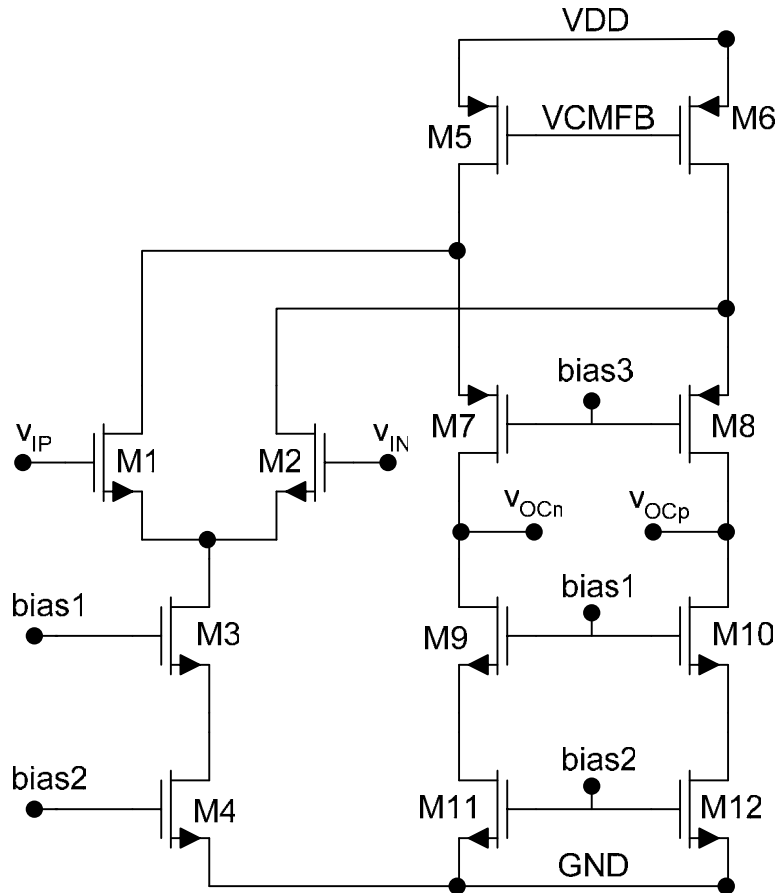
Illustration

- Difficulty in matching P and N boils down to the diagram below
- Note that small differences in current can provide large differences in voltage



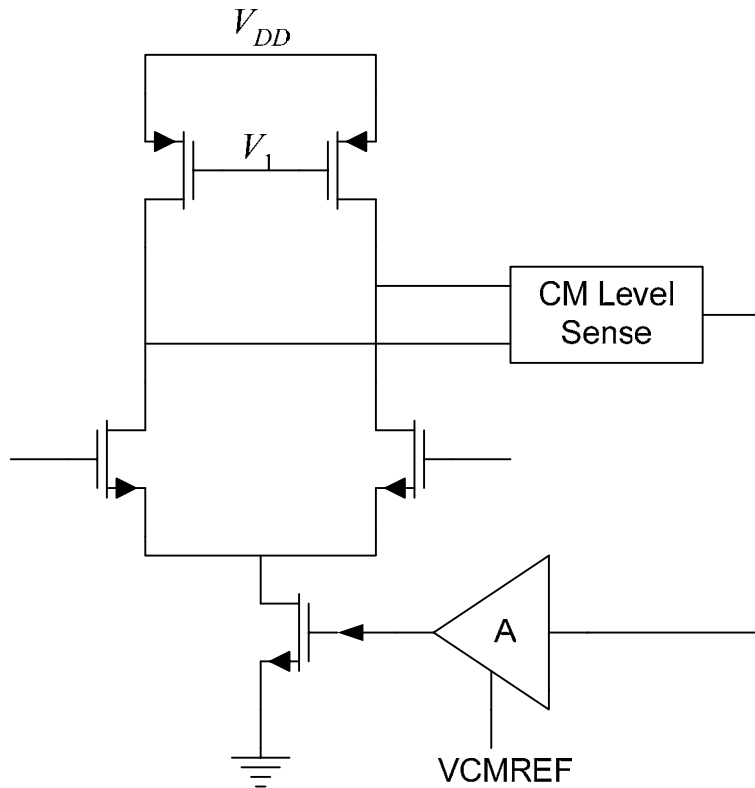
$$\Delta V = \Delta I \times (R_{op} // R_{on})$$

Fully Differential Folded Cascode



- With this configuration there is no way to adjust VCMFB
- A small difference in PMOS vs. NMOS current can cause complete imbalance
- Need a circuit to establish the output common mode
 - ↳ Sense output CM
 - ↳ Adjust output CM by VCMFB
 - ↳ Reject output differential mode

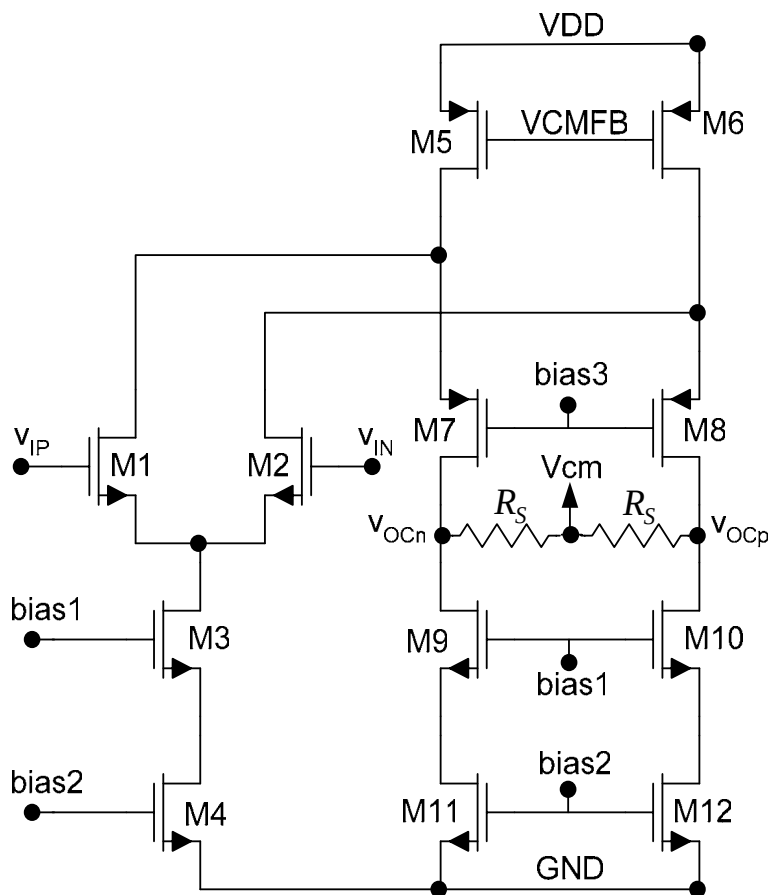
CMFB Topology



- A Common Mode Sense circuit determines the Common Mode Level of the 2 outputs
- This common-mode voltage is compared to a reference
- The error is fed back to the main amplifier's bias network
- Either V_1 or the tail current can be adjusted

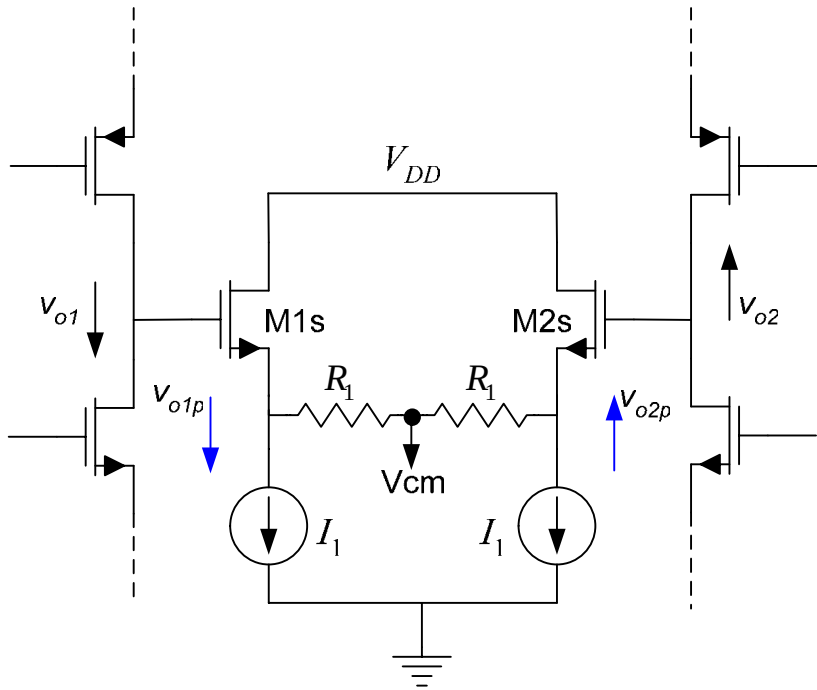
Two Methods of CMFB Exist:
Continuous Time CMFB
Switched Capacitor CMFB

Continuous Time CMFB Topologies



- The sense operation is performed by resistors R_s
- The resistors will need to be very large to maintain the amplifier gain
 - ✚ Large resistors consume large area
 - ✚ They can have substantial parasitics to the substrate
 - ✚ They can lead to noisy performance

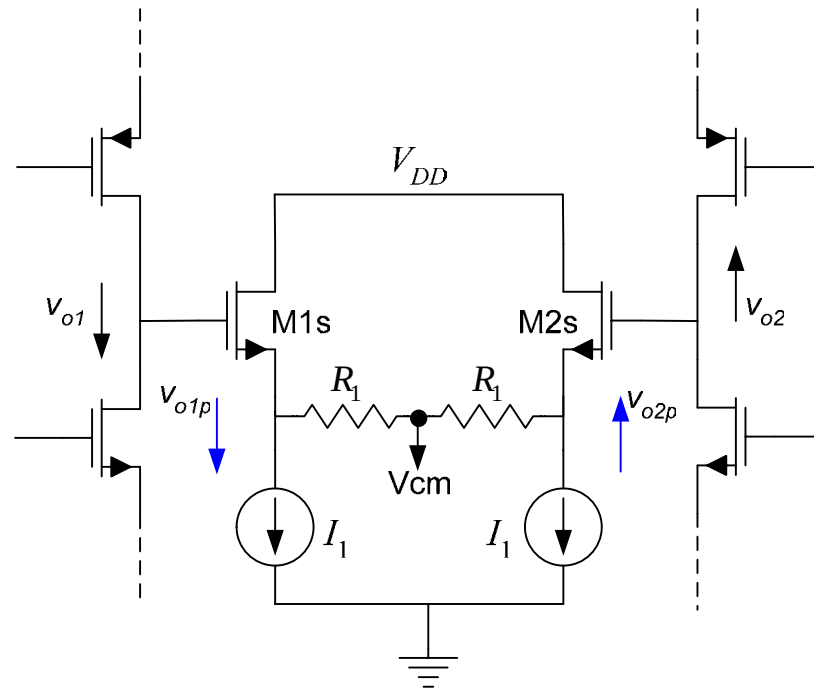
Source Follower Buffering



- M1s and M2s “shield” the output from the loading of the sense resistors
- Actual output common mode levels level shifted by $V_{GS_{M1s,2s}}$
- Must make sure that M1,2s remain in saturation with large differential swings
- Must make sure R1s and I1 current sources are large

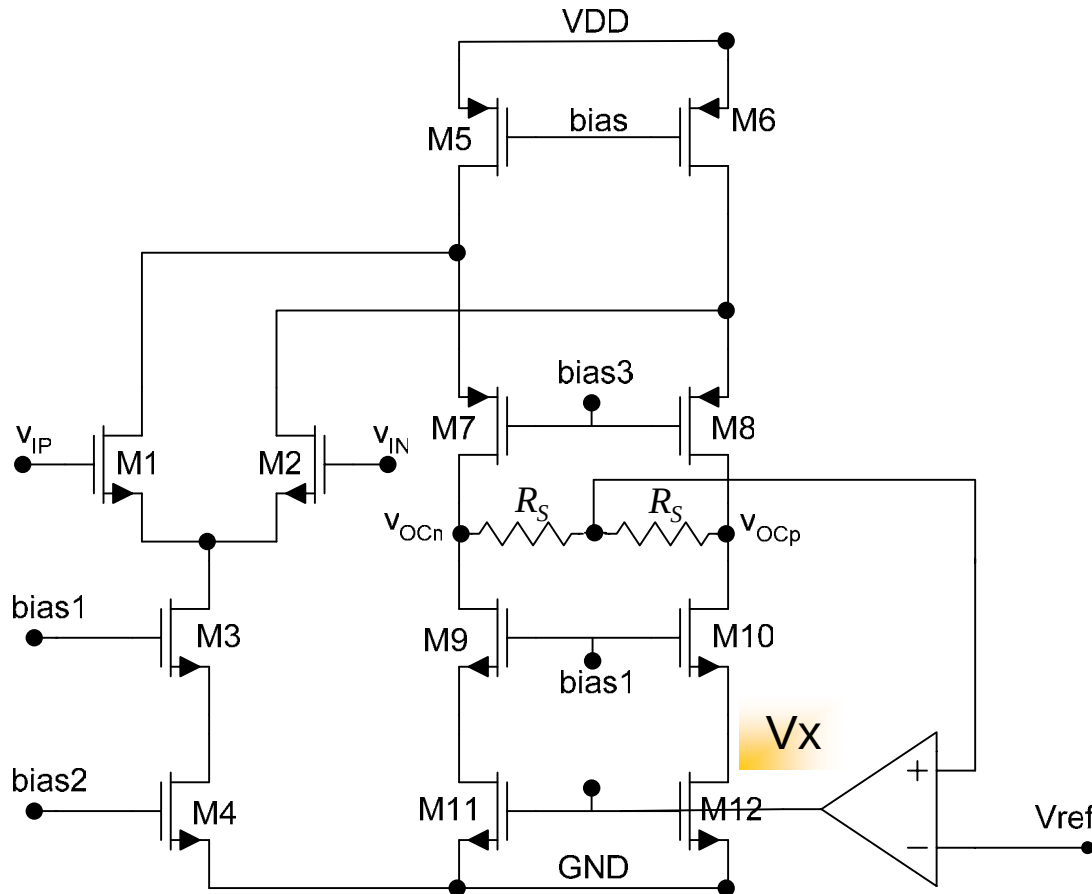
- If the differential swing is large a substantial voltage develops across R1s
 - This can lead to a large current through R1 resistors
- If R1s or I1s are not large enough then either M1s or M2s could turn off disrupting the CM level
- Circuit limits differential output swing even with design precautions in mind

Source Follower Buffering-Swing Implications



- Assume I_1 is derived from a simple current mirror
 - ⇒ The minimum voltage at the drain must be greater than V_{on}
 - ⇒ Add $V_{on} + V_t$ of M1s and the minimum output common mode is $2V_{on} + V_t$
- This is a full V_t worth of output swing removed
- Note that without the source followers the output can swing to $GND + 2V_{on}$

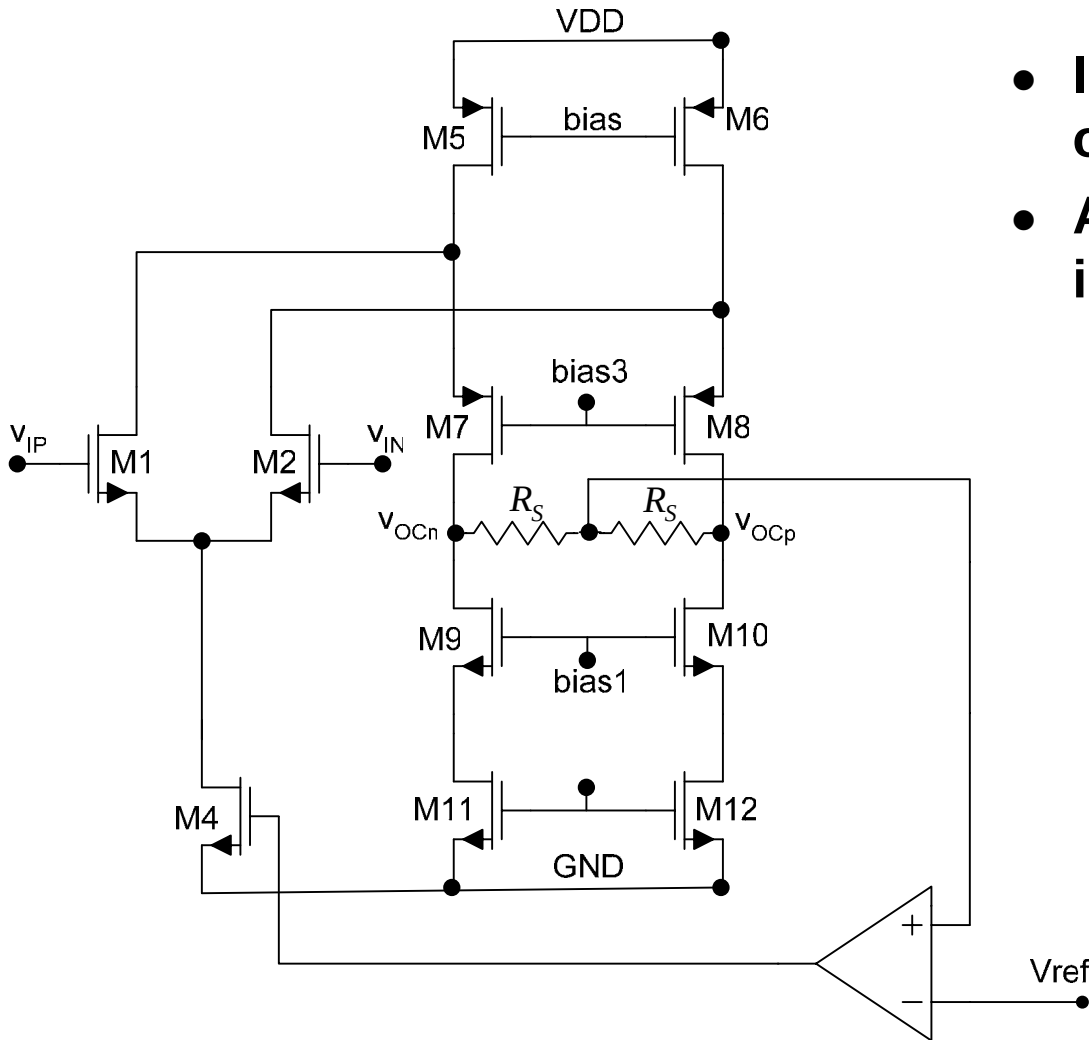
Comparing Measured Common-Mode vs. Reference



- The common mode circuit senses the difference between the output common mode and Vref
- The error is amplified & applied to M11 and M12
- Assume V_{OCn} and V_{OCp} rise in potential
 ➡ Then V_x must follow
- If the gates of M11 and M12 rise in voltage their drain voltage must fall
- This brings V_{OCn} and V_{OCp} down in potential

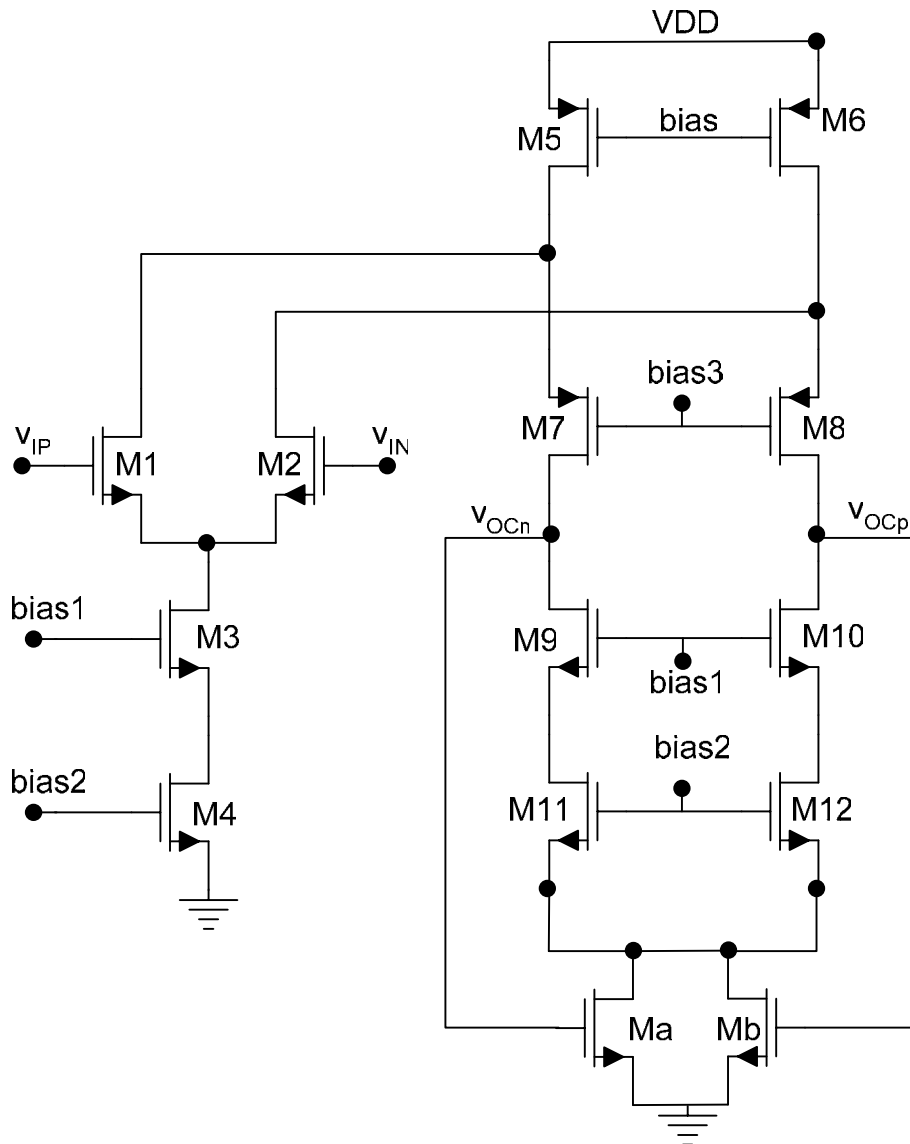
For large loop gain CMFB acts to force the output common mode to Vref

Alternate Control Point



- In this method the tail current of M1 and M2 are adjusted
- Assume V_{ocp} and V_{ocn} rise in potential
 - This will cause the I_{tail} of M1,2 to rise
 - If M1,2 use more current then M7,8 use less
 - This will cause node bias3 to move up in potential (lower V_{GS} for M7,8)
 - If bias3 goes up in potential then V_{ocp} and V_{ocn} will go down

Simple CMFB Topology



- Note that M_a and M_b are in triode
 ↳ The equivalent resistance is

$$R_{eq} = r_{dsa} // r_{dsb}$$

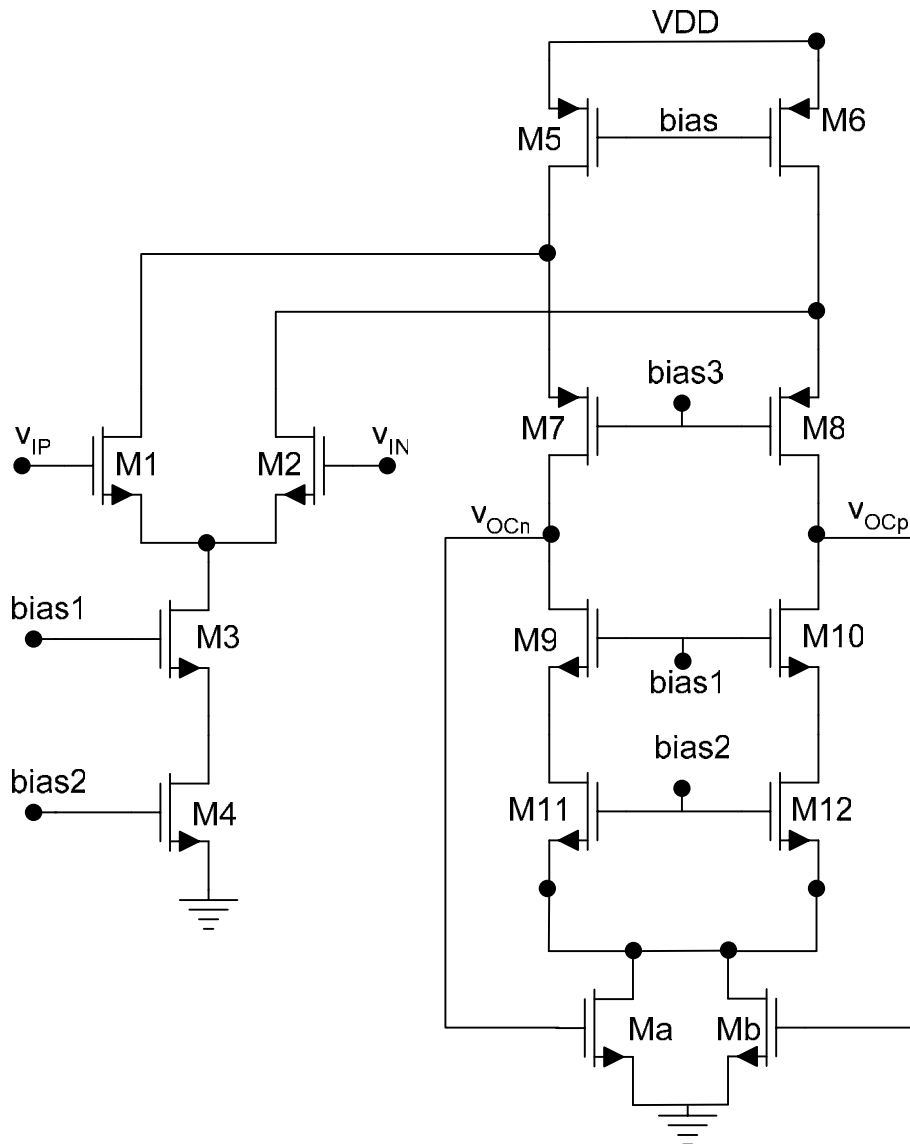
or

$$R_{eq} = \frac{1}{k_n \left(\frac{W}{L} \right) (V_{ocn} - V_t)} // \frac{1}{k_n \left(\frac{W}{L} \right) (V_{ocp} - V_t)}$$

Which results in

$$R_{eq} = \frac{1}{k_n \left(\frac{W}{L} \right) (V_{ocp} + V_{ocn} - 2V_t)}$$

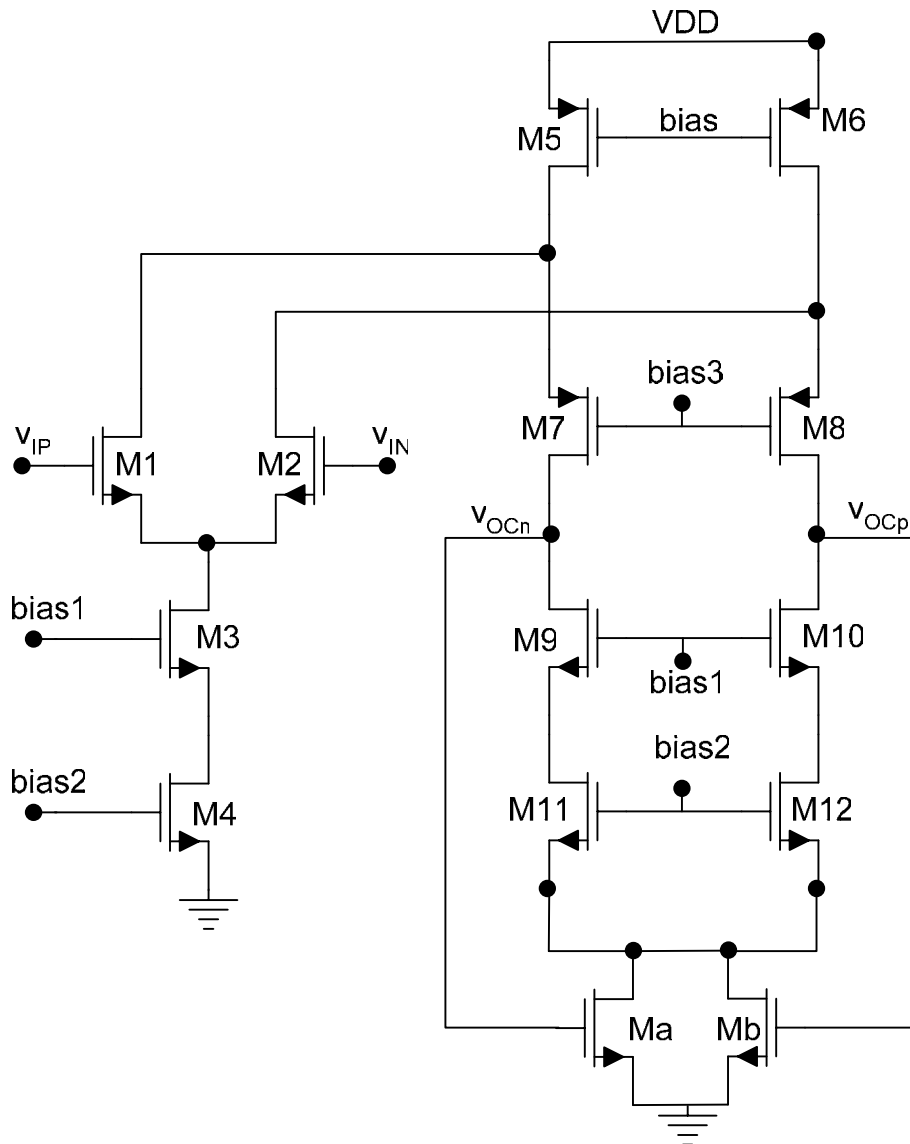
Simple CMFB Operation



- Note that as the output common mode rises as the equivalent resistance of $M_{a,b}$ drops
- If V_{ocn} and V_{ocp} change differentially r_{dsa} and r_{dsb} will take turns increasing and decreasing
- Circuit's benefit is simplicity but it suffers from drawbacks (e.g., extra Clload due to $M_{a,b}$)
- For the CMFB circuit shown the swing is reduced

↪ **Must make sure M_a and M_b remain on**

Simple CMFB Topology Limitations



- Value of CM depends on device parameters
- $M_{a,b}$ will have a finite voltage drop affecting swing
 - ↳ To minimize drop must make $M_{a,b}$ large
- Large $M_{a,b}$ loads the output capacitively

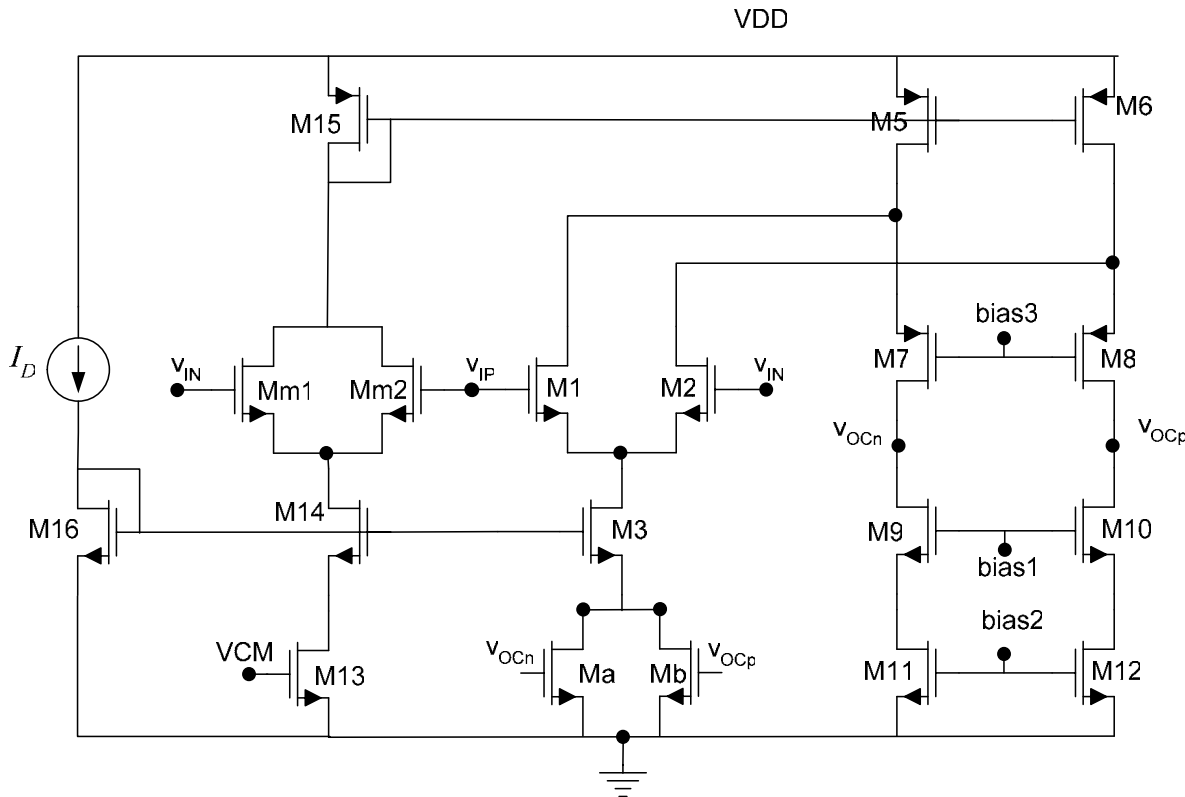
CM dependence on device parameters

$$V_{bias2} - V_{gs11} = 2I R_{eq}$$

$$R_{eq} = \frac{1}{k_n \left(\frac{W}{L} \right) (V_{ocp} + V_{ocn} - 2V_t)}$$

$$V_{ocp} + V_{ocn} = \frac{2I}{k_n \left(\frac{W}{L} \right)} \frac{1}{V_{bias2} - V_{gs11}} + 2V_t$$

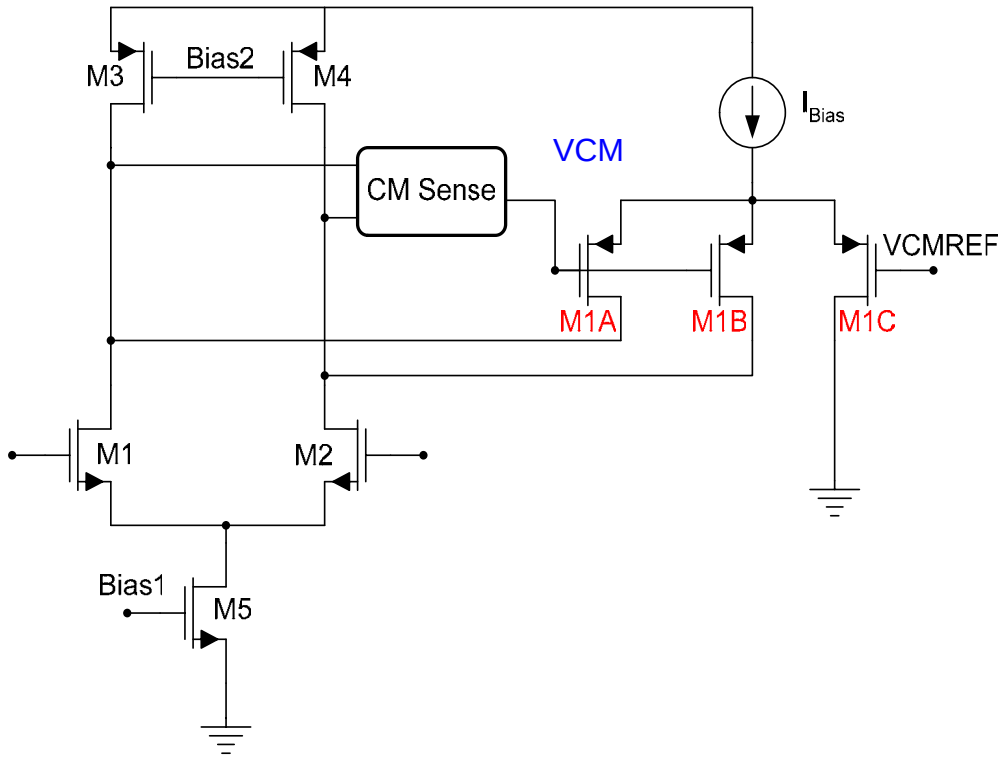
Modified CMFB



- Goal is to make current in M3 track VCM and I_D
 - Assume M16,14 and M3 are matched
 - Assume $(W/L)_{13} = 2(W/L)_{Ma}$
 - Then $I_{D3} = I_D$ only if the output common mode is VCM
 - If V_{ocn} and V_{ocp} go up in potential the CMFB loop will act to bring them back down
 - Ultimate value will be VCM

Transistors Mm1,2 are present to insure V_{DS14} and V_{DS3} track

Alternate CMFB Configuration



$$I_{cm} = \frac{I_{bias}}{4} - \frac{g_{m1A}}{2} (V_{CM} - V_{CMREF})$$

This topology still contributes to output resistive and capacitive loading when connected in this fashion

- **CM sense amplifier directly connects current control to the common-mode outputs**
- **I_{bias} will split between M1A&B and M1C**
- **Note that M5 must now sink The current in the CMFB circuit and inherent current**

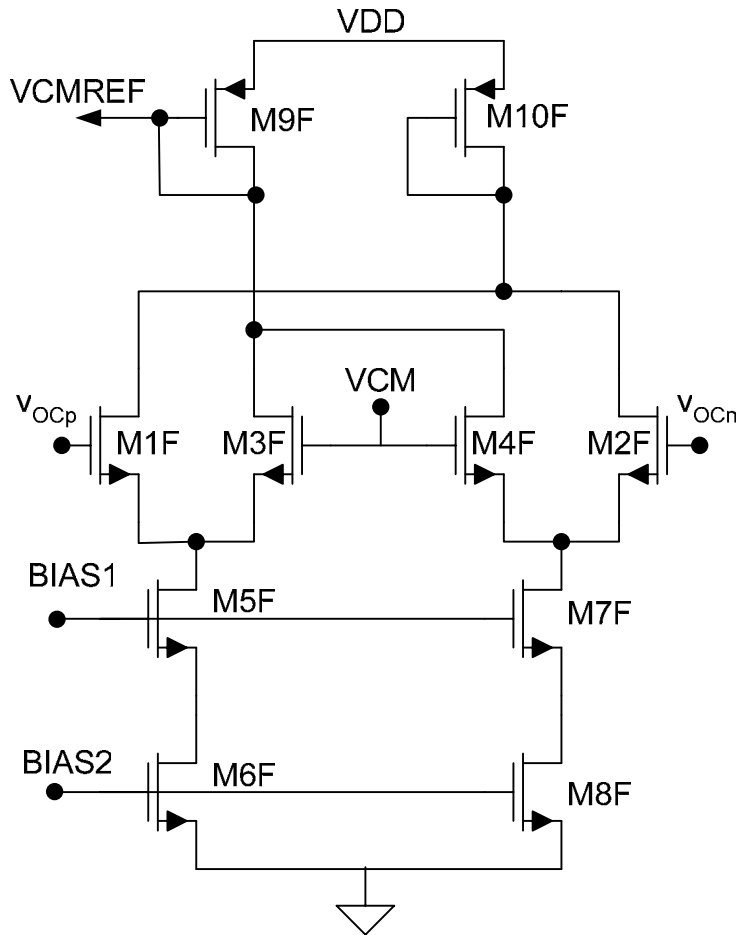
$$I_{D3} + I_{D4} + 2I_{cm} = I_{D5}$$

- Which results in an equivalent current

$$I_{D3} + I_{D4} + \frac{I_{bias}}{2} = I_{D5}$$

- For a cascode circuit CM feedback can be placed at the cascode nodes to avoid output loading

Dual Diff. Pair CMFB



- Assume that V_{ocp} and V_{ocn} are equal and opposite

- As V_{ocp} rises V_{ocn} falls and as a result

$$I_{M1F} = \frac{i_{tail}}{2} + \Delta I$$

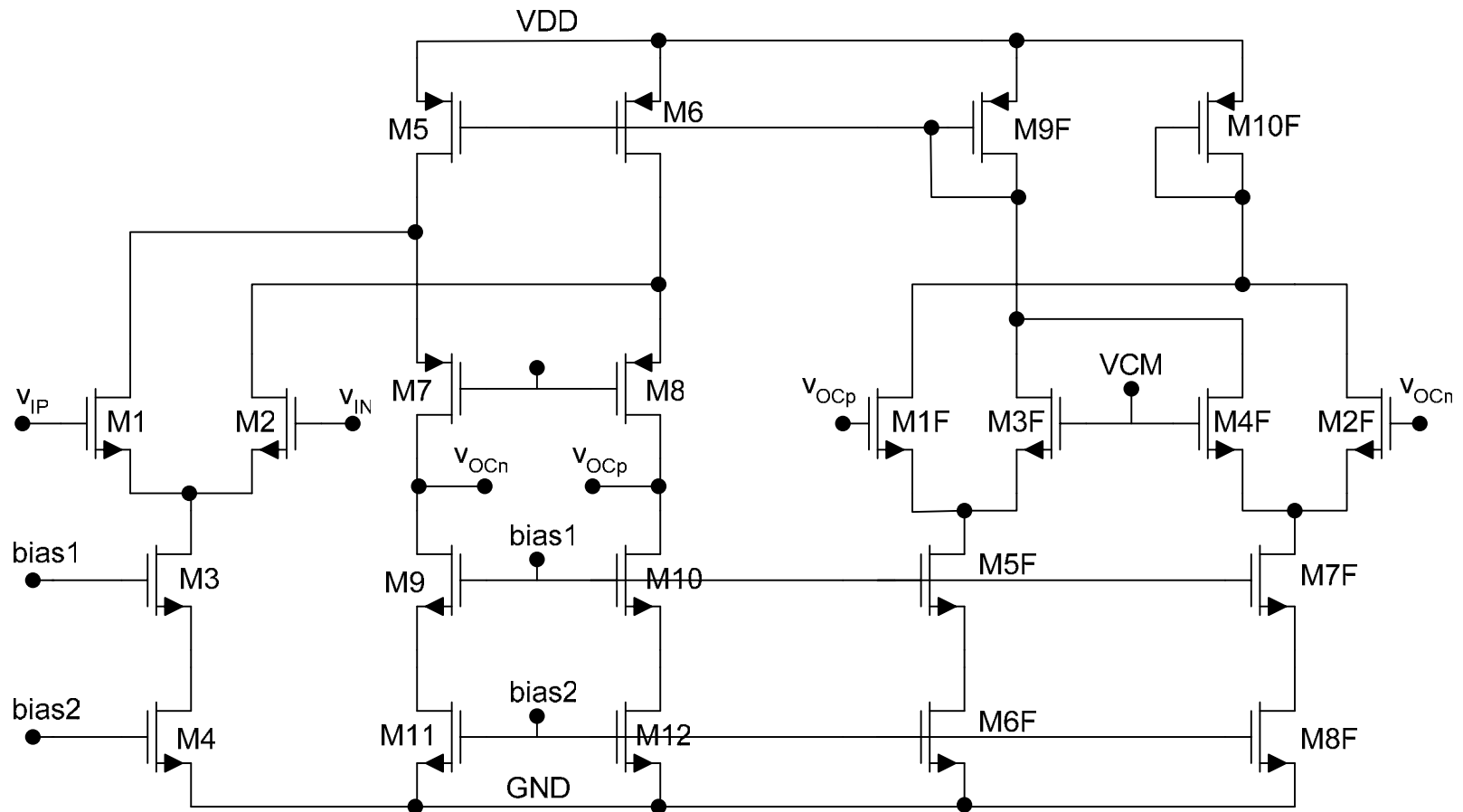
$$I_{M3F} = \frac{i_{tail}}{2} - \Delta I$$

$$I_{M4F} = \frac{i_{tail}}{2} + \Delta I$$

$$I_{M2F} = \frac{i_{tail}}{2} - \Delta I$$

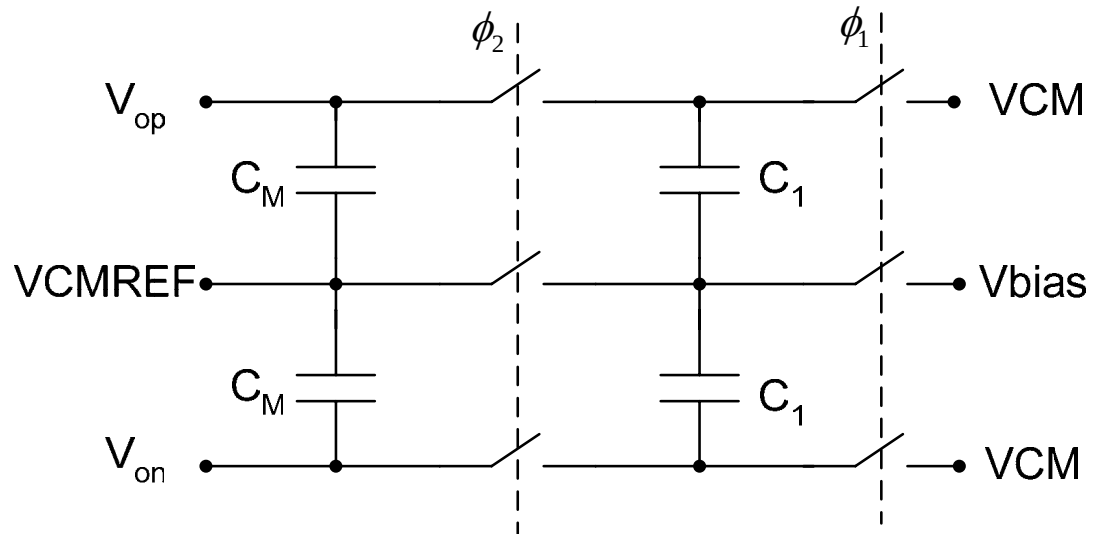
- The current through M9F which sets VCMREF remains unchanged
- Now assume common mode changes
- If the output common mode goes up the currents through M3F and M4F decrease
- As a result VCMREF pulls up and the output pulls down
- The reverse condition follows similar methodology and pulls output CM up

Possible Implementation



Switched Capacitor CMFB

- **Sampled data systems do not use continuous CMFB**
- **Switching noise introduced by sampling events contribute too much noise**
- **SWC CMFB improves on 2 serious limitations**
 - ↳ **Output swing**
 - ↳ **Avoids resistive loading**



- Capacitors C_M generate the average of the output voltages V_{op} and V_{on}
- V_{CMREF} is generated to control the current sources of the amplifier
- The DC voltage of the C_1 capacitors is sampled by capacitors C_M

Switched Capacitor CMFB

- Capacitors C_1 must be of lower size than C_M to avoid excessive loading
 - ↳ The size of C_1 is dependent on the charge injection of the ϕ_1 switch
 - ↳ This charge injection leads to offset
- Use Tgate switches to connect to the output of the opamp for increased swing
- If the tail current is being adjusted with this scheme the tail current transistor can be broken into 2 devices of 50% each (or a more aggressive ratio)
 - ↳ This helps reduce the amount of noise the switch cap circuit injects into the amplifier through the tail current control

